

Phase 4 - Reporting and Insights

Model Comparison

Clustering techniques such as K-Means and Hierarchical Clustering were used to identify patterns in the delivery dataset. The optimal number of clusters was found to be 9, which showed excellent performance with a high Silhouette Score, low Davies-Bouldin Index, and high Calinski-Harabasz Score. These results indicate well-separated and compact clusters.

The neural network model was trained to predict delivery status (Fast or Delayed). After removing data leakage and applying proper preprocessing, the model achieved approximately 80% accuracy on test data. This demonstrates that the model can generalize well, though it depends heavily on the quality of input features.

In comparison, clustering methods were more effective in discovering hidden patterns, while the neural network was useful for prediction. However, clustering provided deeper insights into delivery behavior, making it highly valuable for understanding the system.

Insights from Clustering

Clustering revealed that delivery delays are influenced by multiple factors such as traffic conditions, weather, and distance. High traffic and poor weather conditions were associated with delayed deliveries, while shorter distances and clear weather conditions were linked to faster deliveries.

Certain clusters showed unique patterns, such as long-distance deliveries completed quickly due to efficient routing, and short-distance deliveries delayed due to heavy traffic. This highlights the importance of considering multiple factors together.

Actionable Insights

To improve delivery times, companies should allocate more delivery personnel in high-delay clusters and prioritize orders in high-traffic areas.

Route optimization can be achieved by avoiding routes associated with high-delay clusters and using alternative paths for long-distance deliveries.

Resource management can be enhanced by increasing workforce during peak traffic hours and adverse weather conditions, and reducing idle time in low-demand regions.

Weather-based strategies such as adding buffer time and notifying customers about delays can improve customer satisfaction.